Pranav Nuti

The University of Chicago – Booth School of Business
+1 650 283 9394 • pranav.nuti@chicagobooth.edu
pranavnuti.com

Employment

The University of Chicago Booth School of Business **2024–**
Principal Researcher
Postdoctoral-level position, mentored by Rad Niazadeh

Education

Stanford University **2018–2024**
PhD in Mathematics
Advised by Jan Vondrák

Indian Institute of Science (IISc), Bangalore **2014–2018**
Bachelor of Science (Research), Gold Medalist in Mathematics

Research

Journal Publications and Revisions

Stationary Online Contention Resolution Schemes **Operations Research, Major Revision**
with Mohammad Reza Aminian and Rad Niazadeh
Conference version: EC 2026. Second place, Michael H. Rothkopf Junior Researcher Paper Prize.

Prophet Inequalities with Cancellation Costs **Operations Research, Minor Revision**
with Farbod Ekbatani, Rad Niazadeh, and Jan Vondrák
Conference versions: STOC 2024, EC 2026 (as Optimal Prophet Inequalities with Buyback).

Online Matching and Contention Resolution for Edge Arrivals with Vanishing Probabilities **Operations Research, Minor Revision**
with Will Ma and Calum MacRury
Conference version: EC 2024.

Toward an Optimal Contention Resolution Scheme for Matchings **Mathematical Programming**
with Jan Vondrák
Conference version: IPCO 2023.

The Secretary Problem with Distributions **Mathematics of Operations Research, Major Revision**
Conference version: IPCO 2022.

Peer-Reviewed Conference Publications

Prophet Inequalities and Online Contention Resolution for Matchoids **ITCS 2027**
with Calum MacRury and Jan Vondrák

Single-Sample Prophet Inequalities for Knapsack Problems **ITCS 2027**
with Rebecca Reiffenhäuser and Alice Sayutina

Static Pricing for Single Sample Multi-unit Prophet Inequalities **SOSA 2026**
with Peter Westbrook

Secretary Problems: The Power of a Single Sample **SODA 2023**
with Jan Vondrák

A Tight Competitive Ratio for Online Submodular Welfare Maximization **ESA 2023**
with Amit Ganz and Roy Schwartz

Working Papers

Optimal Contention Resolution from Samples **(submitted)**
with Shaddin Dughmi and Vasilis Livanos

Dynamic Matching with Abandonment: Optimal Guarantees via Static Priority Policies
with Nick Arnosti, Rad Niazadeh, and Felipe Simon

Optimal Bayesian Online Allocation of Reusable Resources
with Mohammad Reza Aminian and Rad Niazadeh

Workshop Papers

Solving Open Problems in Operations Research Using AI **AI-Driven Research in EconCS at EC 2026**
with Eric Fithian and Rad Niazadeh
Draft paper. Based on Open Problems in Operations Research. Also presented as a poster at Can AI do Theory? at STOC 2026.

Bounded Best Buyers: Prophet Inequalities Revisited **Easy Peasy at EC 2026**
with Farbod Ekbatani, Rad Niazadeh, and Jan Vondrák

Magicians Don't Move: An Easy Peasy OCRS **Easy Peasy at EC 2026**
with Mohammad Reza Aminian and Rad Niazadeh

A Match Made by Entropy **Easy Peasy at EC 2026**
with Mohammad Reza Aminian and Rad Niazadeh

Selected Talks

Stationary Online Contention Resolution Schemes **2026**
Young Researchers Workshop, Cornell University

How to Write and Review Papers **2026**
EC 2026 Mentoring Workshop

Prophet Inequalities with Cancellation Costs **2025**
Chicago Junior Theorists Workshop

What is a Good Model for Online Decision Making? **2023**
SURIM Talk, Stanford University

Online Selection Problems and How Ramsey's Theorem Solves a Card Game **2023**
DACO Seminar, ETH Zürich

Between the Secretary Problem and the Prophet Inequality **2022**
Data-Driven Decision Processes Talk, Simons Institute, UC Berkeley

Teaching

Papers

Engaging Activities for Teaching Linear Algebra **PRIMUS 2024**
with Shintaro Fushida-Hardy and Megan Selbach-Allen

Awards

Pólya Teaching Fellow Award **2024**
Awarded for outstanding teaching during the PhD program

Robert Osserman Teaching Award **2022**
Awarded for outstanding contributions as a teaching assistant at Stanford University

Hoefer Mentoring Award **2022**
Awarded for supporting undergraduate writing at Stanford University

Courses Taught as Instructor

Linear Algebra, Stanford Summer Engineering Academy **2021, 2022**
Equity and Inclusion Initiative, Stanford Engineering
This is a summer program that seeks to illuminate the brilliance of students who have been systemically marginalized in engineering. Along with teaching the course, in collaboration with my co-instructors, I also designed the curriculum. 38 classroom hours. Syllabus, lesson plans, handouts and descriptions of all activities are available.

Teaching Outside the University

Stanford Jail and Prison Education Project **2023**
San Francisco County Jail # 5
The project provides education services to people who are incarcerated in Bay Area jails.

International Math Olympiad Training Camp **2015, 2017**
Homi Bhabha Center for Science Education, Mumbai
I taught problem solving, the probabilistic method, elementary number theory, and basic analytic methods in number theory. I also graded exams.

Undergraduate Student Mentorship

Stanford Undergraduate Research in Mathematics **2023**
Department of Mathematics, Stanford University
The program is aimed at introducing Stanford undergraduate students to research in mathematics. I mentored Ethan Zhang, Ezra Steinberg, Peter Westbrook, and Quinn McIntyre. We proved several new results related to prophet inequalities. A report is available.

Directed Reading Program **2020–2022, 2024**
Department of Mathematics, Stanford University
The program is aimed at helping undergraduates, especially members of historically underrepresented groups, participate in the larger mathematical community. I mentored several students: Alessandra Maranca, Sophia Sanchez, Ellen Xu, Donald Poindexter, Logan Bhamidipaty, and Sohan Vichare.

Graduate Student Mentorship

Stanford Summer Engineering Academy **2023**
Equity and Inclusion Initiative, Stanford Engineering
I interviewed instructors to hire for the program, helped design a new curriculum, and provided feedback to instructors.

TA Mentoring Program **2022–2023**
Department of Mathematics, Stanford University
The program is aimed at providing first-time TAs advice and feedback on leading section. I mentored Yosheb Getachew, Joao Campos Vargas, Zhenyuan Zhang, and Hongjian Yang.

Other Awards

Bronze Medal, 54th International Mathematical Olympiad **2013**

KVPY Fellowship **2013–2018**
Awarded by the Govt. of India
A 100% scholarship for undergraduate education.

National Talent Search (NTS) Scholarship **2010–2011**
Awarded by the Govt. of India to high school students